3.3 Quadratic Programming

A convex quadratic with linear constraints. SVM is the running example.

1. Quadratic cost, linear constraints

$$\min_{\boldsymbol{\theta}\in\mathbb{R}^d} \tfrac12\boldsymbol{\theta}^\top Q\boldsymbol{\theta}+\mathbf{c}^\top\boldsymbol{\theta} \quad\text{subject to}\quad A\boldsymbol{\theta}\le\mathbf{b},$$

with Q symmetric **positive definite**. Then the objective is strictly convex, so a local min is the unique global min. Linear constraints keep the feasible set a polyhedron. Soft-margin SVM is exactly this shape.

Ridge regression without inequalities is the unconstrained special case A empty: $\hat{\boldsymbol{\theta}}=-Q^{-1}\mathbf{c}$ if you write the quadratic that way.

---

2. Dual

Lagrangian

$$\mathcal{L}(\boldsymbol{\theta};\boldsymbol{\lambda}) = \tfrac12\boldsymbol{\theta}^\top Q\boldsymbol{\theta}+\mathbf{c}^\top\boldsymbol{\theta}+\boldsymbol{\lambda}^\top(A\boldsymbol{\theta}-\mathbf{b}).$$

Stationarity: $Q\boldsymbol{\theta}+\mathbf{c}+A^\top\boldsymbol{\lambda}=\mathbf{0}$, so $\boldsymbol{\theta}=-Q^{-1}(\mathbf{c}+A^\top\boldsymbol{\lambda})$. Plug back in. The dual is a concave quadratic in $\boldsymbol{\lambda}\ge\mathbf{0}$. You maximize it. Module 3 will do the same substitution for the SVM dual and then replace inner products by a kernel.

---

3. Why SVM is a QP

Primal sketch (soft margin):

$$\min_{\boldsymbol{\theta},\,b,\,\boldsymbol{\xi}} \; \tfrac12|\boldsymbol{\theta}|^2 +C\sum_n\xi_n \quad\text{s.t.}\quad y_n(\mathbf{x}_n^\top\boldsymbol{\theta}+b)\ge 1-\xi_n,\;\xi_n\ge 0.$$

Quadratic objective, linear inequalities. C trades margin size against slack. Week 5 writes this carefully. This week you only need to recognize the type: **QP, not LP**, because of $|\boldsymbol{\theta}|^2$.